Downstream Design Gate: Human Review Before Pilot Stress Maps

SURF2026 American Option Risk Surfaces

Codex-assisted design gate report

June 17, 2026

Abstract

This report translates the completed solver-validation ladder into a practical downstream design policy. Ticket 12 concluded `PASS_SOLVER_VALIDATION_WITH_LIMITATIONS`; the reference integration pass added a shared bibliography. The next step is not large-scale stress maps, datasets, or neural-surrogate labels. The next step is a small human-reviewed pilot that uses the validated baseline American Crank-Nicolson / projected SOR solver, attaches diagnostic metadata to every output, and keeps boundary and Greek caveats visible.

Contents

1 Purpose of the Downstream Design Gate

The project has moved through a validation ladder: Black-Scholes utilities, finite-difference operators, European Crank-Nicolson validation, American CN/PSOR obstacle enforcement, American put and call validation, complementarity diagnostics, threshold-based boundary extraction, Delta/Gamma diagnostics, Rannacher comparison, and grid/domain sensitivity. That ladder is grounded in the classical Black-Scholes and American-option finite-difference setting (Black and Scholes, 1973; Merton, 1973; Brennan and Schwartz, 1977; Wilmott et al., 1995).

This gate answers a different question: how should the validated classical solver be used next? It is a design report, not a new solver ticket. It does not generate stress maps, datasets, neural-network files, or machine-learning labels.

2 Current Validation Status

Ticket 12 made the conservative gate decision

`PASS_SOLVER_VALIDATION_WITH_LIMITATIONS`.

This means the solver is credible enough for human-reviewed pilot design, but not enough for automatic large-scale data generation. The reference integration pass then added verified citations and a shared bibliography, but it did not change the numerical evidence.

Evidence area	Current status	Downstream implication
European pricing	CN matches closed-form prices in the target region.	Price calculations have a trusted benchmark path.
American obstacle	Tested American put and dividend-call cases respect $U \geq \Phi$.	Pilot plots must continue to report obstacle violation.
LCP diagnostics	Equation and complementarity residuals remain near tolerance.	Every pilot run needs LCP diagnostics.
Boundary extraction	Available from continuation premium $U - \Phi$.	Boundary curves are allowed only as threshold-based diagnostics.
Greeks	Delta/Gamma diagnostics exist with masks.	Greeks may be plotted, but not used as labels.
Grid/domain sensitivity	Baseline is usable with cautions.	Pilot defaults are allowed, with selected higher-grid checks.

3 Solver Choice

The default downstream solver is the baseline American CN/PSOR solver created in Ticket 04. The Rannacher-smoothed variant from Ticket 10A remains a comparison tool only. This decision follows Ticket 10A's gate conclusion, `KEEP_BASELINE_FOR_NOW`: prices and boundaries stayed close, but strict-mask Gamma did not improve enough to justify replacing the validated baseline. The project should not silently change the numerical method just because a smoothed variant exists.

Item	Decision	Reason
Default solver	Baseline American CN/PSOR.	It is the solver validated across Tickets 04 through 12.
Rannacher solver	Optional comparison only.	Ticket 10A did not justify adoption as the default.
Smoothing policy	Never hidden.	Any smoothed run must be labeled and compared.
Diagnostics	Required.	PSOR, obstacle, LCP, boundary, and Greek metadata protect interpretation.

4 Grid and Domain Recommendation

For the first pilot, use normalized strike $K = 1$, $S_{\text{max}} = 4K$, and $M = N = 120$. This is the practical default because the validation ladder repeatedly used it, and Ticket 11 showed stable LCP diagnostics and acceptable representative behavior at this size. It is not a production setting.

Before scaling beyond the small pilot, rerun selected pilot cases at $M = N = 180$, still with $S_{\text{max}} = 4K$, as a higher-accuracy check. Interpret the main region $S/K \in [0.4, 1.8]$, but keep the full computational domain for boundary conditions and diagnostics.

Setting	Recommendation	Interpretation
K	1.	Keep normalized strike for pilot readability.
S_{max}	$4K$.	Practical default from validation; larger domains remain checks.
$M = N$	120.	Pilot default balancing runtime and stability.
Higher-grid check	$M = N = 180$.	Confirmation check before scaling, not a proof.
Reported region	$S/K \in [0.4, 1.8]$.	Primary interpretation region from validation plan.

5 Allowed Outputs

The following outputs may be produced in the future pilot if each output carries full parameter metadata and diagnostics. “Allowed” does not mean production-ready; it means acceptable for a small research pilot after human review.

Output	Allowed use	Required metadata
Option value / price	Main pilot price surface and selected slices.	Parameters, grid, convergence status.
Payoff Φ	Benchmark and obstacle reference.	Option type and strike.
Continuation premium $U - \Phi$	Premium surface and boundary input.	Threshold and option-type logic.

Output	Allowed use	Required metadata
Exercise/continuation indicator	Approximate visual region map.	Premium threshold and no-boundary reason where relevant.
Approximate boundary curve	Diagnostic overlay.	Threshold, interpolation rule, and found/not-found status.
PSOR metadata	Required run quality evidence.	Iteration counts, final update, convergence flag.
LCP diagnostics	Required structural evidence.	Obstacle, equation, and complementarity diagnostics.

6 Cautioned Outputs

Boundary and Greek outputs are useful, but easy to overclaim. The free boundary is extracted from the continuation premium $U - \Phi$, so it depends on the grid, threshold, and interpolation rule. That is appropriate for diagnostics, not exact analytical claims.

Delta and Gamma are finite-difference diagnostics. Delta is usually easier to interpret than Gamma. Gamma is especially fragile because it is a second derivative and reacts strongly near payoff kinks, maturity rows, and extracted boundary bands. Rannacher smoothing is a comparison tool for this issue, not the default method (Rannacher, 1984).

Output	Caution	Required reporting rule
Boundary curve	Threshold-based, not exact.	Report threshold, interpolation, and found status.
Exercise indicator	Derived from premium threshold.	Do not describe as exact exercise truth.
Delta	Diagnostic first derivative.	Report endpoint and mask treatment.
Gamma	Fragile second derivative.	Report full-grid and strict-mask values separately.
Greek surfaces	Not labels.	Include kink, maturity, and boundary masks.

7 Blocked Outputs

The following outputs remain blocked by this design gate. They require successful pilot review and a separate approved implementation plan.

Blocked item	Reason
Large training datasets	The project has not yet completed pilot stress-map validation.
Neural surrogate labels	Boundary and Greek outputs remain diagnostic, not label-ready.
Production Greek labels	Gamma remains grid-, kink-, maturity-, and boundary-sensitive.

Blocked item	Reason
Broad parameter sweeps	One-at-a-time pilot checks must come first.
Production risk reporting	The solver is validated with limitations, not certified for production use.
Final paper-style claims beyond evidence	Current evidence does not prove full convergence or exact boundaries.

8 Pilot Stress-Map Proposal

The future pilot should be small enough that a human reviewer can inspect every table and plot. It should not be a dataset-generation job. Use one American put case and one dividend-paying American call case, both with normalized strike $K = 1$.

Case	Base parameters	One-at-a-time variations
American put	$T = 1$, $r = 0.05$, $q = 0.02$, $\sigma = 0.20$, $S_{\max} = 4$, $M = N = 120$.	$\sigma = 0.40$, $r = 0.03$, $q = 0.00$.
Dividend-paying American call	$T = 1$, $r = 0.05$, $q = 0.08$, $\sigma = 0.20$, $S_{\max} = 4$, $M = N = 120$.	$\sigma = 0.40$, $r = 0.03$, $q = 0.03$, $q = 0.10$.

The reported map region should use $S/K \in [0.4, 1.8]$. Time-to-maturity should use $\tau/T \in [0, 1]$, with highlighted slices near 0.01, 0.25, 0.50, 0.75, and 1.00. The solver may compute the full domain, but the report should distinguish full-domain computation from the main interpretation region.

The first plots in the later pilot should be:

- option value surface or heatmap;
- continuation-premium $U - \Phi$ surface or heatmap;
- exercise/continuation indicator map;
- boundary curve overlay on time-to-maturity;
- masked Delta/Gamma diagnostic slices only if mask metadata is included.

9 Acceptance Criteria Before Scaling

The pilot should not scale to broader parameter grids until all of the following checks are met.

Criterion	Required evidence
Solver convergence	All PSOR steps converge, or non-convergence is explicitly reviewed.
Obstacle condition	Maximum obstacle violation remains near zero.
LCP diagnostics	Equation violation and complementarity product remain near diagnostic tolerance.

Criterion	Required evidence
Boundary metadata	Every boundary row records found/not-found status and reason.
Greek masks	Any Greek plot includes kink, maturity, boundary, and strict-mask metadata.
Parameter metadata	Every output records option type, $K, T, r, q, \sigma, S_{\text{max}}, M, N$.
Higher-grid check	Selected cases rerun at $M = N = 180$ before interpretation is broadened.
Human interpretation	Figures are accompanied by caveats and no label-ready claims.

10 Human Review Checklist

Before any future implementation of pilot stress maps, the team or supervisor should answer yes to the following checklist.

Review question	Required answer
Is the baseline CN/PSOR solver accepted as the default?	Yes.
Is Rannacher limited to named comparison runs?	Yes.
Are $K = 1$, $S_{\text{max}} = 4K$, $M = N = 120$, and $M = N = 180$ checks accepted?	Yes.
Are the put and dividend-call pilot parameter ranges reasonable?	Yes.
Are boundary caveats clear enough for readers?	Yes.
Are Gamma caveats and masks required before plotting Greeks?	Yes.
Are datasets, neural labels, and broad sweeps still blocked?	Yes.
Are references and source basis sufficient for a pilot report?	Yes.

11 Risks and Limitations

Risk	Mitigation
Pilot default mistaken for production setting	State that $M = N = 120$ is only the first pilot grid.
Boundary overclaim	Always report threshold and metadata; avoid exact free-boundary language.
Gamma overclaim	Use masks and treat Gamma as diagnostic, not label-ready.
Parameter sweep creep	Use one-at-a-time variations before any grid of regimes.
Dataset creep	Keep CSV/plot outputs as pilot evidence, not training labels.
Hidden solver change	Keep baseline default and label any Rannacher comparison explicitly.

12 Recommended Next Step

The recommended next step is human review of this gate report. If the review accepts the solver, grid defaults, output policy, pilot cases, and blockers, then a later ticket may implement the small pilot stress-map experiment. That later ticket should still be narrow: no large datasets, no neural labels, and no production risk claims.

13 Reproducibility Record

Item	Record
Report source	`reports/04_downstream/downstream_design_gate.tex`.
Compiled report	`reports/04_downstream/downstream_design_gate.pdf`.
Policy table	`reports/04_downstream/downstream_design_policy.csv`.
Bibliography	`reports/03_solver/references.bib`.
No code changes	This gate creates report and policy files only.

References

Fischer Black and Myron Scholes. The pricing of options and corporate liabilities. *Journal of Political Economy*, 81(3):637–654, 1973. doi: 10.1086/260062.

Michael J. Brennan and Eduardo S. Schwartz. The valuation of american put options. *The Journal of Finance*, 32(2):449, 1977. doi: 10.2307/2326779.

Robert C. Merton. Theory of rational option pricing. *The Bell Journal of Economics and Management Science*, 4(1):141, 1973. doi: 10.2307/3003143.

Rolf Rannacher. Finite element solution of diffusion problems with irregular data. *Numerische Mathematik*, 43(2):309–327, 1984. doi: 10.1007/BF01390130.

Paul Wilmott, Sam Howison, and Jeff Dewynne. *The Mathematics of Financial Derivatives*. Cambridge University Press, 1995. doi: 10.1017/CBO9780511812545.